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COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, FLIGHT LINE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-T-83808 for all federal agencies.

1. SCOPE

This commercial item description (CID) covers the requirements for the Flight Line Tow Tractor (FLTT), hereafter referred to as FLTT. The FLTT is based on a commercial truck chassis, also known as a chassis-cab, which is modified for use as a FLTT in worldwide service. The FLTT is capable of towing trailers weighing up to 40,000 pounds and military aircraft weighing up to 85,000 pounds on grades up to 2.5% on United States Air Force installations.

2. CLASSIFICATION

2.1 FLTT classes. The two classes specified shall be of the same design, with the only differences being in the chassis components necessary for a four-wheel drive FLTT. The tow tractor shall be one of the following classes specified:

Class 1	two-wheel drive FLTT (standard equipment)
Class 2	four-wheel drive FLTT (optional equipment, see 7.3)

Beneficial comments, recommendations, additions, deletions, clarifications, and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Wright-Patterson AFB, OH 45433-7017 or email to engineering.standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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3. SALIENT CHARACTERISTICS

3.1 FLTT description. The FLTT is based on a truck chassis, which is modified for use as a FLTT in worldwide service. The Air Force (AF) uses the FLTT to tow Aircraft Ground Equipment (AGE), munitions trailers, and military aircraft. The modifications to the commercial truck chassis include shortening the length of the chassis by cutting the frame rails and shortening the wheelbase by relocating the rear axle, suspension, and other affected components forward on the original chassis frame rails. Other modifications include the addition of a ballast body over the rear axle for traction, a tool box, and hitches for towing.

3.2 Design and construction. The design shall promote cost-effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. The FLTT shall be designed and manufactured in accordance with Society of Automotive Engineers (SAE) specifications ARP1247 and AIR 1363. The FLTT shall be designed and constructed so that no parts will work loose in service and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. Location of Center of Gravity of ballast body shall not adversely impact the ability to make front-end alignment adjustments within the factory camber-adjustment range. Each completed FLTT shall undergo a front-end alignment to factory alignment specifications.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used.

3.2.1.2 Finish. The exterior finish color of the FLTT shall be the FLTT manufacturer's standard commercial color. When specified (see 7.3), the exterior finish color of the FLTT shall be the FLTT manufacturer's commercial gloss white. Also when specified (see 7.3), the exterior finish color of the FLTT shall be Coast Guard Blue, color number 15182 of SAE AMS-STD-595. Prior to the application of the finish top coat, any component or assembly specifically manufactured for or by the FLTT manufacturer, from sheet, plate, angular, or tubular steel, shall be coated with a zinc-rich primer. Prior to applying any refinish primer and top coat to the chassis-cab original finish, the original finish shall be prepared for recoating by scuffing with an abrasive material and cleaning at a minimum. The refinish top coat shall extend into the door jambs and hood jambs to the extent that the Original Equipment Manufacturer (OEM) finish color cannot be seen through the gaps between the body panels when the doors and hood are closed.

3.2.1.3 Exclusion of water. The design of the FLTT shall be such as to prevent water leaking into, or being driven into, any part of the FLTT interior when stationary or moving. All windows, doors, panels, covers, or related type items, shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to

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prevent wetting of equipment and heat- and sound-proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the FLTT that could collect and retain water.

3.2.1.3.1 Fluid traps and faying surfaces. There shall be no fluid traps on the FLTT. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.3.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.3.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall permit drainage of all fluids. The minimum size of the drain holes shall be 0.25-inch.

3.2.1.4 Corrosion protection. The FLTT shall be corrosion-protected with a permanent, black, commercially-available corrosion-preventive compound (CPC) that is difficult to remove (CORTEC VpCI 375 or equivalent). Application shall include the exposed chassis-frame metal underneath the FLTT (excluding the drive train), the inside of the wheel wells, and behind the fender skirts. Any equivalents used shall meet the below salient characteristics in table I when applied on AISI 1010 carbon steel Q-panels:

TABLE I. VpCI 375 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	4.0-4.5	2500+
ASTM D1748	4.0-4.5	2000+
ASTM D870	4.0	312+

In addition to the equivalent requirements in table I, the pencil hardness in accordance with American Society for Testing and Materials (ASTM) D3363 shall be “H”, the temperature resistance shall be -150 to 350 °F, and the adhesion in accordance with ASTM D3359 shall be “5B”. When specified (see 7.3), FLTTs that are to be shipped overseas shall be temporarily corrosion-protected from the effect of salt spray both on the ship and at port with the application of a thin, soft-film, semi-transparent, commercially available CPC that is easily removable by washing or with a low pressure steam (Cortec VpCI 389 or equivalent). Application of the temporary CPC shall be to the exterior of the FLTT, in critical or corrosion sensitive areas of the FLTT. Any equivalents used shall meet the below salient characteristics in table II when applied on AISI 1010 carbon steel:

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TABLE II. VpCI 389 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	2.0	600+
ASTM D1748	1.0	1000+
ASTM G85 (94)	1.0	600+

3.2.2 Markings. All external devices that require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1.00-inch-high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Black Int'l, Navy #3 Black, ANA 604, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Red Int'l, CARC Aircraft Red, ANA 619, and Color Number 31136 of SAE AMS-STD-595. Each side of the FLTT shall be marked with a center-of-gravity decal applied within 1.00 inch of the calculated center of gravity. Tire inflation pressures shall be marked above all wheel well openings, with 0.50 inch high letters/numbers that are black in color. A decal shall be included on the front and rear of the vehicle near the pintle-hook position that states the maximum towing capacity in terms of drawbar pull force (DBPF) of the FLTT and Gross Vehicle Weight (GVW) of the towed trailers and aircraft. The decal shall read as follows:

MAXIMUM TOWING CAPACITY: 4,500 LBS DRAWBAR PULL FORCE / 40,000 LBS GVW TRAILERS / 85,000 LBS GVW AIRCRAFT

3.2.3 Identification and data plates.

3.2.3.1 Identification plate. The FLTT shall be provided with a permanently marked identification plate constructed of a non-corroding metal mounted on the inside of the FLTT cab on the driver door panel and a rear license plate mounting bracket with license-plate attaching hardware. The variable information on the identification plate shall be engraved onto the plate by a method that produces a consistent lettering quality. At a minimum, the identification plate shall include the following information:

a. Vehicle nomenclature.

b. Manufacturer's part number and/or model number.

c. Vehicle identification (serial) number (VIN).

d. Date of manufacture.

e. Manufacturer's name, address, and phone number.

f. Commercial and government entity (CAGE) code.

g. National Stock Number (NSN).

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h. Contract number.

i. Registration number.

j. Date of warranty expiration.

The invariable information contained on the identification plates shall be documented in the technical manual. The FLTT shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate, marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human-readable markings.

3.2.3.2 Transportation data plate. The FLTT shall be provided with a permanently marked transportation data plate constructed of a non-corroding metal mounted on the exterior surface of the FLTT. The plate shall contain at least the following information:

a. Side and rear silhouette views of the FLTT.

b. Horizontal and vertical location of the center of gravity of the FLTT in air-transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

The information contained on the transportation data plate shall be documented in the technical manual.

3.2.3.3 Lubrication data plate. The FLTT shall be provided with a permanently marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the FLTT that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 3.3.1. The information contained on the lubrication data plate shall be documented in the technical manual.

3.2.3.4 OEM data plates. Data (such as tire air pressures, chassis wheelbase, GVW, etc.) found on the factory-original commercial truck chassis data plates (normally found in the door jambs) that is inconsistent with the modified commercial truck chassis design shall be replaced with data pertaining to the new design. Decal overlays are acceptable.

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3.2.4 Environment, Safety, and Occupational Health (ESOH).

3.2.4.1 Component protection. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.2 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the FLTT with wire ropes or chains. "Dog tag"-style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve-stem caps shall be made of plastic.

3.2.4.3 Fire extinguishers. The FLTT shall be equipped with two 5-pound (minimum) dry-chemical type fire extinguishers with mounting brackets suitable for use on vehicles and with Underwriters Laboratories (UL) rating of 20B:C (minimum). The fire extinguishers shall be mounted either on the top surface of the ballast body behind the cab, or on the rear surface of the ballast body, and accessible from the driver and passenger side.

3.2.4.4 Back-up alarm. An externally mounted back-up warning alarm shall be included in the reverse-light circuit.

3.2.5 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Tapped threads shall have a minimum thread engagement in accordance with table III.

TABLE III. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.2.6 Welders and welding. Welding and post-heat treatment procedures shall be in accordance with the American Welding Society (AWS) welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.2.7 Special tools. The design of the item shall minimize the requirement for special tools (see 7.5.2). All special tools shall be provided with, and stored on, the FLTT.

3.2.8 Cab specifications.

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3.2.8.1 Cab design. The FLTT shall be equipped with a three-man cab. Interior cab sound level shall not exceed 84 dB (A) with doors and windows closed. Seating shall be covered in cloth. The first entrance step to the cab shall be a full-length, running-board-style step and shall not exceed 18.00 inches in height from the ground. The outer edge of this running-board-style step shall be offset out from the outer edge of the cab door opening by at least 4.75 inches. Cab instrumentation shall feature an hour meter in addition to the standard instrumentation. When an air system for towing trailers is specified (see 3.2.10), cab instrumentation shall include an air-pressure gauge and a low-pressure indicator with buzzer and/or warning light. The FLTT shall have a fixed, recessed compartment for tools with a hinged door/lid and latching mechanism located behind the cab in an easily accessible location. The door/lid shall be supported when opened, and the latching mechanism shall not bind when the door/lid is closed. The recessed compartment shall be a minimum of 36.00 inches in length, 12.00 inches in width, and 9.00 inches in depth, and shall be constructed from a minimum of 11 gauge steel. The FLTT shall be equipped with an air conditioning system. The FLTT cab shall be equipped with wipers, a windshield washer, driver- and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The cab shall feature a 12V auxiliary power outlet and a rear-sliding window.

3.2.8.2 Cab visibility. The FLTT shall be designed to provide maximum visibility in all directions for the range of persons represented between the 5th-percentile female driver and the 95th-percentile male driver. The three rear pintle hooks shall be visible either directly or via mirrors from the driver position. If mirrors are required, one mirror per pintle hook shall be provided to complete this visibility requirement. Mirrors used for pintle-hook visibility shall be convex with a convex radius between 20.00 and 25.00 inches. A plane defined by a line 18.00 inches above the rear pintle hooks and a point represented by the driver's eye shall not be obstructed by any structure or feature on the truck. The light bar is exempt from this requirement.

3.2.8.3 Cab noise levels. Interior cab sound level shall not exceed 84 decibel adjusted (dB(A)) with doors and windows closed. If the sound level is above 84 dB(A), the contractor shall provide a hazardous noise warning plate stating that operator's hearing protection is required when operating this FLTT. The plate shall be made of a corrosion resistant material and shall be at least 4.00 square inches. It shall have black lettering on a yellow background and shall be located to be visible as the operator enters the cab, or is seated. Sound level shall not exceed 94 dB(A).

3.2.9 Wheels and tires. The FLTT shall be equipped with steel disc wheels and tubeless steel belted radial tires with on/off-road tread and. When the class 2 FLTT is specified (see 7.3), the FLTT shall be equipped with non-directional mud and snow tires. The valve stems may be rubber or metal. Tire valve-stem caps shall be plastic. The wheels and tires shall be in accordance with the *Tire and Rim Association* recommendations for the application. All wheels and tires shall be uniform. If tire loads and tire inflation pressures that exceed the specifications on the tire sidewall markings are required, a copy of the application approval from the tire manufacturer shall be documented in the technical manual.

3.2.10 Air system. When specified (see 7.3), the FLTT shall have an air system for towing trailers with air brakes. The air system shall be an over-the-road commercial truck/tractor design to include a reservoir tank with heated and automatic drain valves, automotive compressor, dryer, trailer

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protection valve, low-pressure indicator, service and emergency glad hands, glad-hand covers with lanyards, and any other necessary components. The glad hands shall be in accordance with SAE J318, and shall be polarized and identified as “service” and “emergency”. The glad hands shall be mounted on the passenger side rear surface of the FLTT, within 26.00 inches (horizontal distance) of the centerline of the vehicle, within 12.00 inches (vertical distance) above the top of the rear-center pintle hook while in its uppermost height position, and within 15.00 inches forward of the base of the rear-center pintle hook. Components of the air system that are sensitive to the elements (rain water, road water, ice build-up, road grit, etc.) and installed in positions on the FLTT that are vulnerable to those elements shall be covered or shielded to prevent the premature failure of those components over time or during service.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The FLTT shall be capable of operating in ambient temperatures ranging from –20 to 125 °F. The FLTT shall be capable of operating in ambient temperatures to –40 °F when heavy-duty (type A) winterization system is specified (see 7.3). Any requirements for towing in high ambient temperatures (recommended coolant ratios, oil types and viscosities, or other fluid types, for various temperature ranges) shall appear in the technical manual.

3.3.2 Storage temperature. The FLTT shall be capable of being stored in ambient temperature to –65 °F when heavy-duty (type A) winterization system is specified (see 7.3).

3.4 Gross vehicle weight (GVW) and dimensions.

3.4.1 GVW. The FLTT shall have a minimum GVW of 8,000 pounds. The GVW shall not exceed gross vehicle weight rating (GVWR) minus 1,500 pounds. The minimum GVWR required for this application is 9,500 pounds.

3.4.2 Dimensions. Overall dimensions shall not exceed:

Length: 220.00 inches, without pintle hooks

Width: 100.00 inches, without mirrors

Height: 100.00 inches

3.5 Transportability.

3.5.1 Air transportability. The FLTT shall be transportable on C-130, C-17, and C-5 aircraft in accordance with MIL-STD-1791. In all air-transport configurations, the FLTT shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The FLTT shall be equipped with pressure-relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.

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3.5.1.1 Shoring. The FLTT shall drive on and off the aircraft, negotiating the required maximum ramp angles without approach and step-up shoring. The FLTT shall be transported on the aircraft without load shoring, parking shoring, and sleeper shoring.

3.5.1.2 Axle weight. Single-axle weight shall not exceed 13,000 pounds.

3.5.1.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) for air transport.

3.5.1.4 Air-transportable configuration. The air-transportable configuration shall include all equipment, ½ tank of fuel, and no crewmembers. At least 6.00 inches sidewall and 6.00 inches overhead clearance shall be maintained between the FLTT and the aircraft at all times during loading and flight. The restrained FLTT shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

3.5.1.5 Equipment removal and reconfiguration. Preparation and restoration of the FLTT for air transportability shall take no more than 30 minutes for two persons using common, non-powered, hand tools. All equipment removed shall be stored on the FLTT; caps and plugs shall permit driving and storage in transport configuration.

3.5.2 Tie downs. The FLTT shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the FLTT. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.5.3 Lifting provisions. The FLTT shall be equipped with sufficient attachment points so located that it can be lifted by crane; each attachment point shall be marked "Lift Point". The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

3.6 Performance.

3.6.1 Mobility. The FLTT shall be capable of 37 to 40 miles per hour (MPH) (minimum) on flat asphalt or concrete surfaces without a trailing load, and shall be governed not to exceed 40 mph (plus 0, minus 3). Reverse speed shall be a minimum of 2.5 mph on flat surfaces. The FLTT shall be capable of towing trailers from a stop to 10 mph (minimum) and aircraft from a stop to 5 mph (minimum) on grades up to 2.5% on asphalt or concrete surfaces. The FLTT shall be capable of operating on 10-percent side slopes and 12-percent grades without fluid spillage.

3.6.2 Drawbar pull force (DBPF). The FLTT shall have the following minimum DBPF of 4,500 pounds on dry, level, concrete (coefficient of friction of 0.8).

3.6.3 Chassis modifications.

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3.6.3.1 Chassis frame modifications. The truck-chassis wheelbase shall be modified to 124.00 inches or less. The rear axle, suspension, and other affected components shall be relocated forward on the original chassis-frame rails. Relocated fasteners shall be preloaded in parallel tension and shall not be subject to bending stress due to chassis-frame surface irregularities. Optimal angles and smooth operation shall be preserved in the driveline. Modifications to chassis components shall use an engineering design approach and quality workmanship to preserve structural integrity. Excess length at the rear of the chassis-frame rails shall be removed using a process that leaves a smooth-cut finish. Raw metal exposed by the modifications shall be primed and painted chassis color.

3.6.3.2 Chassis fuel-system modifications. Fuel fill or supply hoses shall not be exposed. Splicing into the OEM metal fuel lines with flexible fuel-lines is permitted. Flexible fuel lines shall be pressure rated for fuel-injection systems. Fuel hose clamps shall be rated for fuel-injection systems or shall be constant-torque, worm-drive hose clamps. Modifications to the OEM fuel lines shall withstand fuel-system pressure. Optionally, OEM metal fuel lines may be cut, and a flare and tube nut may be added. Female-inverted, "field-attachable" brass or steel fittings may be attached to the integral tube nut. Field-attachable fittings do not require a hose clamp when the applicable flexible hoses and correct working pressures are selected.

3.6.4 Driveline and steering. A class 1 FLTT shall be offered as a two-wheel drive chassis. When specified (see 7.3), a class 2 FLTT shall be offered in a part-time, four-wheel drive chassis. The axle-reduction ratio shall be the same on the class 1 and 2 FLTT. The FLTT shall have a Single Rear Wheel (SRW) axle with a limited-slip differential (preferred) or an electronic locking differential (alternate) and shall have a reduction ratio equal to or greater than 3.73:1. The two-wheel drive and four-wheel drive FLTT shall have a maximum wall-to-wall turning radius of 23.50 feet or less. The wheelbase shall be 124.00 inches or less.

3.6.5 Brakes.

3.6.5.1 Parking brake. The FLTT shall be equipped with a parking brake that shall be in compliance with SAE J293. The FLTT manufacturer shall ensure that any parking-brake cable relocation or cable shortening (resulting from modifying/shortening the wheelbase of the commercial truck chassis) is equivalent to the original chassis-cab manufacturer's design in providing parking brake cable tension.

3.6.5.2 Service brakes. The FLTT shall feature a braking system in compliance with Federal Motor Vehicle Safety Standards (FMVSS) for vehicles of this GVWR at the time of manufacture.

3.6.6 Pintle hooks. The FLTT shall be equipped with four pintle hooks that incorporate a cotter or snapper pin as a secondary locking device and shall conform to CID A-A-52550, type 1, class 2. Pintle hooks shall be attached to the pintle hook mounting structure with a minimum of four ½-13 or ½-20, grade 8 fasteners. The pintle hook fasteners shall be installed with a design fastening torque applied such that an adequate preload is introduced in the fastener to prevent losing or fatiguing the fasteners and/or the pintle hook. A description of the pintle hook fastener type and the fastening torque shall be recorded in the technical manual.

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3.6.6.1 Front pintle-hook position. The front of the FLTT shall be equipped with one center-mounted pintle hook, with the coupling position of the pintle hook located at 18.50 inches from the ground, plus or minus 0.50 inch. The front pintle-hook assembly and mounting structure (all structural components inclusive between the pintle hook and frame) shall be designed to withstand the following load forces multiplied by the respective Safety Factors (SF) and shall be the minimum loads for attaining design stress levels:

a. A downward vertical force value (tongue load) of 1000 pounds multiplied by a SF of 1.0

b. A tension and compression force value of 5,000 pounds applied in the same vector direction as DBPF in a straight pull or push, multiplied by a SF of 2.0

3.6.6.2 Rear-outboard pintle hook positions. The rear of the FLTT shall be equipped with two outboard pintle hooks located in the same vertical plane, with the pintle hook centerlines at 43.00 inches to each side of the FLTT centerline, with the coupling positions of the pintle hooks located at 18.50 inches, plus or minus 0.50 inch, from the ground. The rear-outboard pintle-hook assemblies and mounting structures (all structural components inclusive between the pintle hooks and frame) shall be designed to withstand the following load forces multiplied by the respective Safety Factors (SF), and shall be the minimum loads for attaining design stress levels:

a. A downward vertical force value (tongue load) of 300 pounds multiplied by a SF of 1.0

b. A tension and compression force value of 5,000 pounds applied in the same vector direction as DBPF in a straight pull or push, multiplied by a SF of 1.0

3.6.6.3 Rear-center pintle hook position. The rear of the FLTT shall be equipped with one center-mounted, extendable pintle hook with pintle-hook coupling positions that are height-adjustable from 12.00 inches to 18.50 inches from the ground. A minimum of four evenly spaced or coupling positions (including 12.00 inches and 18.50 inches) shall be provided. Height adjustment shall be easily made by the operator without the use of tools. The mounting surface of the rear-center extendable pintle hook, when retracted for towing, shall not extend more than 5.00 inches behind the mounting surfaces of the rear-outboard pintle hooks. The extendable unit shall consist of a self-locking sliding bar that shall allow the hook backward movement of not less than 10.50 inches, or more than 12.00 inches, and a lateral-swing movement of 45 degrees in either direction from the centerline of the FLTT. The unit shall be fitted with rollers or low-friction pads, and provisions shall be made for lubricating all moving parts. Low-friction pads shall be positively secured such that repeated lateral movements over the life of the vehicle do not result in premature pad, bonding adhesive, or fastener failure. The normal towing position shall be with the pintle-hook/hitch assembly in the retracted and straight positions, rather than the extended and swung positions. The extendable bar shall be equipped with a safety grab handle, attached to the pintle-hook/hitch assembly, and positioned above the pintle hook, to allow for easy user access and function of the extendable hitch front-to-rear and side-to-side hitch motion, as well as the pintle hook height adjustments. The extendable bar shall be easily removable to facilitate inspection and maintenance. The extendable/swing features are required to facilitate connection of the pintle hook to the trailer by a single operator, rather than a method requiring two users. Once the lunette or towing eye of the towed load is secured, the extendable unit shall be engaged in the straight and closed positions for towing by the forward movement of the FLTT until the hitch assembly is straight and the self-

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locking device is secure, followed by the backward movement of the FLTT until the hitch assembly is closed and the self-locking device is secure. A note shall be added in the technical manual that states the following: “Nuclear towing is authorized only when the rear, center pintle hook is locked in the closed position, both fully retracted and centered. Positioning and locking of hitch assembly by towing or backing the FLTT is not authorized.” Any shear pins, locking pins, or related pintle hook components that are designed to be removed for adjustments by the user shall be attached to the FLTT with lanyards. Removable items that are heavier (such as the pintle hook assembly if applicable) shall be attached with an up-sized, heavier gauge lanyard. Shear pins that use a locking pin for retention shall use a locking pin that is a double-loop hair-pin design. If constructed of mild steel, the hitch shall have hardened steel bushings and pins at all pivot and locking points. The bushings shall be easily removed for service. The rear-center pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall be designed to withstand the following load forces multiplied by the respective Safety Factors (SF), and shall be the minimum loads for attaining design stress levels:

In the extended (open) position:

a. A downward vertical force value (tongue load) of 1000 pounds multiplied by a SF of 1.0

b. A tension and compression force value of 5,000 pounds applied in the same vector direction as DBPF in a straight pull or push, multiplied by a SF of 1.0

In the retracted (closed) position:

a. A downward vertical force value (tongue load) of 1,000 pounds multiplied by a SF of 1.0

b. A tension force value of 5,250 pounds applied in the same vector direction as DBPF and a compression force value of 6,500 pounds in the same vector direction as DBPF in a straight pull or push, multiplied by a SF of 2.0

c. A tension and compression force value of 500 pounds applied at 45 degrees from the same vector direction as DBPF in a straight pull or push, multiplied by a SF of 3.0

3.7 Engine and related equipment.

3.7.1 Engine. The FLTT shall have a commercially available, liquid-cooled, turbocharged diesel engine of not less than six cylinders and power train sufficient to develop the required drawbar pull and obtain the required FLTT speeds. The diesel engine shall operate on high-sulfur diesel fuels (containing over 15 parts per million (ppm) sulfur), and ultra-low sulfur diesel fuels (containing less than 15 ppm sulfur). At a minimum, the FLTT shall meet a Tier II level exhaust and evaporative emissions standard (see 7.6.1) for on-highway engines. The engine shall include an anti-restart, as well as a neutral safety start feature.

3.7.2 Engine starting aids. The engine shall start within 15 seconds cranking in any ambient temperature within the required operating range of the FLTT. Internal engine starting aids, fluid

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starting aids, and heat from the winterization system (see 3.7.6) may be used prior to and during the start period to facilitate engine starting under the following conditions in table IV:

TABLE IV. Engine starting aids.

Temperature Range	Starting Aids Permitted
40 through 125 °F	None
0 through 39 °F	Internal engine starting aids and fluid starting aids
–40 through –1 °F	Internal engine starting aids, fluid starting aids, and heat from the winterization system

3.7.3 Engine cooling system. The FLTT engine shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of 125 °F without exceeding the manufacturer's maximum allowable temperatures on the individual engine components.

3.7.4 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. The exhaust system outlet shall exit on the right-hand side, over the rear axle and behind the rear wheel, oriented at 90 degrees from the centerline of the FLTT, and not directed toward the ground.

3.7.5 Transmission. Transmission shall be of the automatic-shift type. Maximum allowable transmission temperature during all operations shall not exceed 260 °F.

3.7.6 Winterization system. A standard (type C) winterization system shall be provided for operation in temperatures to -20 °F. When specified (see 7.3), a heavy-duty (type A) winterization system shall be provided for operation to -40 °F and storage to -65 °F. The winterization systems may include heaters for engine coolant, engine oil, fuel, and battery warmers. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. For each winterization system, a breakdown of the winterization components and requirements (recommended coolant ratios, oil types and viscosities, or other fluid types, for the various temperature ranges) shall be documented in the technical manual.

3.7.6.1 Winterization plugs and connectors. Winterization package components with external power requirements shall be compatible with 110-volt (V) alternating current (AC), 60 hertz (Hz) power source using covered male inlet plugs in accordance with National Electrical Manufacturers Association (NEMA) WD 6.

3.8 Electrical system. The FLTT shall be equipped with a 12V, negative-ground electrical system. All external chassis wiring shall incorporate weather-proof connections and terminations. Exceptions to this are under-hood terminal strips and electrical posts that are protected from the elements. Any wire splicing (typically resulting from special-request options) shall incorporate a

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mechanically-crimped connector, solder, and heat-shrink tubing. Crimped-on wire connectors shall be installed with a corresponding crimping tool. All wiring shall be sized to handle the system voltage and current levels. Battery cables shall be identified with "+" and "-" or red and black markings, and shall not be spliced.

3.8.1 Lighting. The FLTT lighting shall be equivalent to the standard commercial offering to include a rear adjustable-aim floodlight, parking lights, head lights (high and low beam), brake lights, turn signals, hazard lights wired to a constant-hot circuit, and reverse lights. A flashing amber beacon light shall be mounted above the roofline on the FLTT centerline. A switch shall be mounted inside the cabin within reach of the seated operator for actuating the beacon light. Two double-faced signal lamps (SAE J588) shall be mounted above the roofline, 48.00 inches apart (24.00 inches off FLTT centerline) and shall be integrated with the FLTT taillights. The flashing beacon light and double-faced signal lamps shall be mounted on a light bar that is not fastened to the cab that runs approximately 1 inch above the roofline to avoid drilling and cutting into the cab. The rear adjustable-aim floodlight, of not less than 3000 candlepower, shall be mounted on the rear edge of the ballast body so that the rear coupling positions are illuminated. The FLTT shall be equipped with an adjustable spotlight, capable of illuminating 180 degrees of elevation and 360 degrees of rotation, and located on the driver side of the cab. The spotlight shall be installed in the driver door-to-windshield pillar and shall be steerable from the driver position inside the cab. The specified lights added by the FLTT manufacturer shall be constructed with Light-Emitting Diodes (LED).

3.8.2 Batteries and battery compartment.

3.8.2.1 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top-post type in accordance with MIL-B-18013/1.

3.8.2.2 Battery compartment. The batteries shall remain in the OEM location and configuration.

3.8.3 12-volt trailer-light harness receptacle. The rear of the FLTT shall be equipped with an SAE J560 electrical receptacle, mounted on the driver side rear surface of the FLTT, within 24.00 inches (horizontal distance) of the centerline of the vehicle, within 12.00 inches (vertical distance) above the top of the rear-center pintle hook while in its uppermost height position, and within 15.00 inches forward of the base of the rear-center pintle hook. All terminals of the SAE J560 electrical receptacle shall be wired in accordance with SAE J560. A quantity of six (6) relays shall be incorporated to protect and isolate the FLTT OEM chassis wiring from the load demands of terminals #2 through #7 in the SAE J560 receptacle. Terminals #2 through #6 shall be supplied with a constant-hot battery source. Terminal #7 shall be supplied with an ignition-hot battery source. Each circuit within the SAE J560 receptacle shall have a minimum circuit protection of 20 amps. The relays and circuit-protection devices shall be mounted on a panel or panels and installed in a practical location on the FLTT.

3.9 Towing safety attachments.

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3.9.1 Breakaway cable hookup. The rear of the FLTT shall be equipped with an emergency trailer-breakaway-cable attachment point, capable of mating with a National Aerospace Standards (NAS) 1281C10 spring snap hook. The attachment point shall be located on the FLTT centerline, within 8.00 inches (vertical distance) above the top of the center pintle hook while in its uppermost height position, and within 4.00 inches forward of the base of the pintle hook. The attachment point shall not be installed on the pintle-hook assembly.

3.9.2 Safety-chain attachment points. The rear of the FLTT shall have two safety-chain attachment points for use when towing, composed of hinged D-rings suspended in the downward position. The D-rings shall be located symmetrically on each side of the center pintle hook at a distance of 26.00 inches above the ground. The vertical centerline of the D-rings shall be located at a parallel distance of not greater than 6.00 inches to the left and right of the FLTT centerline. The rear edge of the D-rings shall be within 2.00 inches forward of the base of the pintle hook. The D-rings shall not be installed on the pintle-hook assembly. Each D-ring shall be capable of withstanding a maximum shock load of 6,000 pounds and shall have the ability for standard safety-chain hooks to be attached. The inside radius of the safety-chain hook is between 1.00 and 3.00 inches.

3.10 Workmanship. The FLTT, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.10.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.10.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.10.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.10.4 Cleaning. The FLTT shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.11 Cybersecurity. The FLTT shall be designed, produced, and delivered with cybersecurity considerations IAW DoDI 8500.01, DoDI 8510.01, and AFI 17-101 and referenced security controls for applicable Information Technology (IT) and/or Operational Technology (OT), commonly referred to as Platform Information Technology (PIT). Commercial policy and standards, such as SAE J3061, SAE J3101, and various ISO/IEC standards may be used in lieu of

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DoDI/AFI, as applicable. The safety of the driver, passengers, and bystanders shall be the most important security consideration.

3.11.1 Secure Communication. The FLTT shall be designed, produced, and delivered with the bare minimum of any wireless communication (i.e. cellular, Wi-Fi, telematics, over-the-air (OTA) updates, etc.) while still meeting operational/performance specifications within this CID. All attempts shall be made to utilize OEM commercial basic ordering features (i.e. lack of options) to meet the intent of this requirement without physical modification to commercial vehicles that could adversely affect vehicle functionality via internal communication or void any standard commercial warranty. Any wireless communications remaining within the final configuration of the FLTT shall be secured to mitigate known vulnerabilities and threats IAW applicable policy and standards as referenced in paragraph 3.11. All risks associated with known vulnerabilities and potential threats of any wireless communication capability shall be identified, assessed, and reported.

3.11.2 Hardware Supply Chain Integrity. The FLTT shall be designed, produced, and delivered with security controls for the supply chain integrity of electronics hardware (i.e. engine control units (ECUs)) that involve memory storage, data processing, and/or interfaces with the communication system or controller area network (CAN bus) within the vehicle. The FLTT shall be produced with mitigation to the risk of counterfeit electronic parts being installed in end items or otherwise entering the United States Air Force (USAF)/Department of Defense (DoD) inventory.

3.11.3 Software/Firmware Supply Chain Integrity. The FLTT shall be designed, produced, and delivered with security controls for the supply chain integrity of all delivered software and firmware. All software and firmware shall function as intended, and shall be free of vulnerabilities (i.e. defects, bugs, malware, etc.), either intentionally or unintentionally inserted as part of the software/firmware throughout the lifecycle. Any diagnostic and troubleshooting software or utility software for updating firmware shall be compatible for use on a Microsoft Windows 10 (64-bit) operating system computer. Any interface hardware/cable (i.e. CAN bus, USB, RS-232) required to communicate between a host computer with utility/diagnostic software and the FLTT shall be provided with each vehicle.

4. **REGULATORY REQUIREMENTS**

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life-cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list

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that contains recovered materials, unless the item cannot be acquired a) competitively within a reasonable timeframe, b) meeting appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. **PRODUCT CONFORMANCE PROVISIONS**

The products provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Pre-production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 Pre-production inspection. The pre-production FLTT shall be subjected to demonstrations, examinations, and tests, and supplied with analyses, certifications, lists of components, and schematics, as described in 5.5.1 through 5.5.28.2. The contractor shall provide or arrange for all test equipment and facilities.

5.3 Conformance inspection. Each production FLTT shall be subjected to the examination described in 5.5.1.

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data shall be recorded. Additional data shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the FLTT shall be closely observed for the following conditions, which shall be cause for rejection:

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a. Failure to conform to design or performance requirements specified herein.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the OEM specifications or tolerances shall be used for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the FLTT components or between the FLTT, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

(1) Engine cooling system.

(2) Engine lubrication system.

(3) Engine protective circuits.

5.5 Test methods.

5.5.1 Examination of product. Each FLTT shall be examined to verify compliance with the requirements. The pre-production FLTT (see 5.2) shall be examined using a contractor-generated, Government-approved test procedure to identify each requirement verified by demonstration, examination, or test. Attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each FLTT function shall be verified. Analyses, certifications, lists of components, and schematics shall be provided in accordance with table V. Each production FLTT (see 5.3) shall be examined using a contractor-generated, Government-approved checklist that is a reduced version of the pre-production checklist.

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TABLE V. Analyses, Certifications, Lists of Components, and Schematics.

Paragraph	Required Analyses, Certifications, Lists of Components, and Schematics
3.2 Design and construction.	Contractor certification (see 5.5.2)
3.3.1 Operating temperature range.	Contractor certification (see 5.5.11.1)
3.3.2 Storage temperature.	Contractor certification (see 5.5.11.2)
3.5.1 Air transportability.	Contractor air transportability analysis (see 5.5.13.1)
3.5.1.1 Shoring.	
3.5.1.4 Air transportable configuration.	
3.5.2 Tie downs.	Contractor tie down provision analysis (see 5.5.13.3)
3.5.3 Lifting provisions.	Contractor lifting provision analysis (see 5.5.13.4)
3.6.4 Driveline and steering (axle).	Axle certification (see 5.5.18.1)
3.6.5.1 Parking brake.	Contractor certification (see 5.5.19)
3.6.6 Pintle Hooks.	Contractor analysis (see 5.5.20)
3.7.1 Engine.	Engine certification (see 5.5.21)
3.7.3 Engine cooling system.	
3.7.6 Winterization system.	Contractor list of components and requirements (see 5.5.23)
3.8.3 12-volt trailer light harness receptacle.	Contractor electrical schematic (see 5.5.27)
3.11 Cybersecurity	Contractor certification and analysis (see 5.5.29)

5.5.2 Design and construction certification. A contractor certification shall be submitted to the Government representative at the pre-production inspection stating that the FLTT has been designed and constructed per SAE specifications ARP1247 and AIR 1363, in compliance with the requirements of 3.2. The contractor certification shall include a front-end alignment report showing the camber values are within the original chassis-cab manufacturer's alignment specifications, in compliance with the requirements of 3.2.

5.5.3 Finish examination. The pre-production FLTT shall be examined for verification of the paint-finish top coat, in compliance with the requirements of 3.2.1.2.

5.5.4 Corrosion protection examination. The pre-production FLTT shall be examined for verification of the permanent CPC, in compliance with the requirements of 3.2.1.4.

5.5.5 Markings examination. The pre-production FLTT shall be examined for verification of markings, in compliance with the requirements of 3.2.2.

5.5.6 Identification and data-plates examination. The pre-production FLTT shall be examined for verification of the identification and data plates, in compliance with the requirements of 3.2.3.1, 3.2.3.2, 3.2.3.3, and 3.2.3.4.

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5.5.7 Fire extinguisher examination. The pre-production FLTT shall be examined for verification of the fire extinguisher, in compliance with the requirements of 3.2.4.3.

5.5.8 Back-up alarm examination. The pre-production FLTT shall be examined for verification of the back-up alarm, in compliance with the requirements of 3.2.4.4.

5.5.9 Cab verification.

5.5.9.1 Cab design examination. The pre-production FLTT shall be examined for verification of cab design, in compliance with the requirements of 3.2.8.1.

5.5.9.2 Cab visibility examination and test. The pre-production FLTT shall be examined and measured for verification of cab visibility, in compliance with the requirements of 3.2.8.2.

5.5.9.3 Cab noise level test. The pre-production FLTT shall be tested by measuring the noise level at the operator's ear with the engine operating at maximum RPM and windows and doors closed, for verification of compliance with the requirements of 3.2.8.3. If noise level is between 84 and 94 dB(A), the noise warning plate required by 3.2.8.3 shall be on all FLTTs delivered on this contract.

5.5.10 Tires and wheels examination. The pre-production FLTT shall be examined for verification of tires and wheels, in compliance with the requirements of 3.2.9.

5.5.11 Environmental conditions verification.

5.5.11.1 Operating temperature range certification. A contractor certification shall be submitted to the Government representative at the pre-production inspection stating that the FLTT and all temperature-sensitive components are capable of operation in the specified temperature ranges, in compliance with the requirements of 3.3.1.

5.5.11.2. Storage temperature certification. A contractor certification shall be submitted to the Government representative at the pre-production inspection stating that the FLTT and all temperature-sensitive components are capable of storage to the specified temperature, in compliance with the requirements of 3.3.2.

5.5.12 Weight and dimension tests.

5.5.12.1 Weight test. The pre-production FLTT shall be weighed for verification of the GVW and axle weights, in compliance with the GVW requirement of 3.4.1 and the single-axle weight requirement of 3.5.1.2.

5.5.12.2 Dimension measurement test. The pre-production FLTT shall be measured for verification of the dimensions, in compliance with the requirements of 3.4.2.

5.5.13 Transportability verification.

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5.5.13.1 Air transportability analysis. An engineering analysis shall be performed for verification of compliance with the air-transportability requirements of 3.5.1, 3.5.1.1, and 3.5.1.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.5.1. The engineering analysis shall also include a dimensional analysis for the FLTT while traversing the ramp and while loaded aboard C-130, C-17, and C-5 aircraft.

5.5.13.2 Equipment removal and reconfiguration demonstration. The pre-production FLTT shall be configured for transport on C-17 and C-5 aircraft and then reconfigured for operation, for verification of compliance with the requirements 3.5.1.5. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

5.5.13.3 Tie down provision analysis. An engineering analysis shall be performed and submitted to the Government representative at the pre-production inspection, for verification of compliance with the tie down provision requirements of 3.5.2.

5.5.13.4 Lifting provision analysis. An engineering analysis shall be performed and submitted to the Government representative at the pre-production inspection, for verification of compliance with the lifting provision requirements of 3.5.3.

5.5.14 Mobility tests. The pre-production FLTT shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the FLTT speedometer as the speed-measuring device, in compliance with the requirements of 3.6.1.

5.5.15 Drawbar pull test. The pre-production FLTT shall be tested for verification of the DBPF on dry, level concrete using a load-sensing device and a fixed anchoring position, in compliance with the requirements of 3.6.2. The drawbar position shall be horizontal and level.

5.5.16 Chassis frame modifications. The pre-production FLTT shall be examined for verification of workmanship of chassis frame modifications, in compliance with the requirements of 3.6.3.1.

5.5.17 Chassis fuel-system modifications. The pre-production FLTT shall be examined for verification of workmanship of chassis fuel-system modifications, in compliance with the requirements of 3.6.3.2.

5.5.18 Driveline and steering verification.

5.5.18.1 Axle certification. A manufacturer certification shall be submitted to the Government representative at the pre-production inspection stating that the axle has a limited-slip differential (preferred) or an electronic locking differential (alternate), and a reduction ratio equal to or greater than 3.73:1, for verification of compliance with the requirements of 3.6.4.

5.5.18.2 Driveline and steering test. The pre-production FLTT shall be tested for verification of compliance with the requirements of 3.6.4. The FLTT shall be tested for verification of tire-chain use without interference in wheel wells. Tire chains shall be fitted to the tires, and the FLTT shall be operated by driving a short distance and turning the wheels to the extreme left and right. The FLTT shall be tested for verification of the maximum wall-to-wall turning radius while in four-

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wheel coordinated steering mode. The fully ballasted pre-production FLTT shall be tested as follows for verification of compliance with the turning-radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the FLTT being tested.

b. Equipment required.

(1) A device suitable for measuring three times the length of the FLTT being tested with an accuracy of at least ±1.00 inch.

(2) Markers or marking device suitable for marking the pavement.

(3) A plumb bob suitable for locating a point on the pavement directly below a fixed point on the FLTT.

(4) A test-report notebook or similar record forms to be used as a test-report worksheet and incorporated into the documentation package.

c. Test Conditions.

(1) The FLTT shall be fully ballasted and in operational form.

(2) The FLTT steering system shall be fully operational, and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.

d. Test Procedure.

(1) The FLTT shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.

(2) Continue driving the slow, full cramp circle.

(3) At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the FLTT using the service brakes.

(4) At each stop, place a plumb bob against the outermost point of the FLTT and mark the spot on the ground directly below where the plumb bob comes to rest.

(5) Measure and record the straight-line distances between each pair of points, that is, Lengths AB, BC, and CA.

(6) Calculate the wall-to-wall turning radius (R) as follows:

R = D / 2 where

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$$D = \frac{(AB \times BC \times CA)}{2 \times \sqrt{(S \times (S - AB) \times (S - BC) \times (S - CA))}} \text{ and}$$

$$S = (AB + BC + CA) / 2$$

e. Repeat steps (1) through (6) with the FLTT moving in the opposite direction.

5.5.19 Parking-brake certification and examination. A contractor certification shall be submitted to the Government representative at the pre-production inspection stating that the parking brake on FLTT is equivalent to the original chassis-cab manufacturer's design in providing parking-brake cable tension, in compliance with the requirements of 3.6.5.1. The pre-production FLTT shall be examined to ensure that any parking-brake cable relocation or cable shortening (resulting from modifying/shortening the wheelbase of the commercial truck chassis) is equivalent to the original chassis-cab manufacturer's design in providing parking-brake cable tension, in compliance with the requirements of 3.6.5.1.

5.5.20 Pintle hooks and mounting structures analysis and examination. An engineering analysis report shall be performed and submitted to the Government representative at the pre-production inspection, for verification of compliance with the strength requirements of 3.6.6. The report shall contain supporting calculations to show that the front and rear pintle-hook assemblies and mounting structures are in compliance with their respective strength specifications. The calculations shall be supplemented with a Finite Element Analysis (FEA) to further support compliance with the strength specifications. The pre-production FLTT shall be examined for verification of the specified pintle hooks and locations, in compliance with the requirements of 3.6.6.

5.5.21 Engine and related equipment.

5.5.21.1 Engine and engine cooling system certification. A manufacturer certification shall be submitted to the Government representative at the pre-production inspection stating that the engine is in compliance with the requirements of 3.7.1, including exhaust-emissions standards and fuels, and 3.7.3.

5.5.21.2 Transmission. The pre-production FLTT shall be examined for verification of the automatic-shift transmission, in compliance with the requirements of 3.7.5.

5.5.22 Exhaust-system examination. The pre-production FLTT shall be examined for verification of the exhaust system, in compliance with the requirements of 3.7.4.

5.5.23 Winterization system, list of components and system requirements, and examination. A list of components for the winterization system and requirements (recommended coolant ratios, oil types and viscosities, or other fluid types, for the various temperature ranges) for each winterization system shall be submitted to the Government representative at the pre-production inspection. Reference the requirements of 3.7.6. The pre-production FLTT shall be examined for

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a winterization system (either type C or type A) in compliance with the requirements of 3.7.6 and 3.7.6.1.

5.5.24 Electrical system examination. The pre-production FLTT shall be examined for verification of an electrical system, in compliance with the requirements of 3.8.

5.5.25 Lighting examination. The pre-production FLTT shall be examined for verification of lighting, in compliance with the requirements of 3.8.1.

5.5.26 Batteries and battery-compartment examination. The pre-production FLTT shall be examined for verification of batteries, in compliance with the requirements of 3.8.2.1 and 3.8.2.2.

5.5.27 12-volt trailer light-harness receptacle schematic, examination, and test. An electrical schematic of the 12-volt trailer light harness circuitry and its interface with the OEM wiring shall be submitted to the Government representative at the pre-production inspection, for verification of compliance with the requirements of 3.8.3. The pre-production FLTT shall be examined for verification of the 12-volt trailer light harness receptacle, in compliance with the requirements of 3.8.3. Each circuit in the receptacle shall be tested with a light tester or a voltmeter to confirm circuit function when actuated, in compliance with the requirements of 3.8.3.

5.5.28 Towing safety attachments.

5.5.28.1 Breakaway cable-hookup examination. The pre-production FLTT shall be examined for verification of the breakaway cable hookup and location, in compliance with the requirements of 3.9.1.

5.5.28.2 Safety-chain attachment points examination. The pre-production FLTT shall be examined for verification of the safety-chain attachment points and location, in compliance with the requirements of 3.9.2.

5.5.29 Cybersecurity Analysis. An engineering analysis shall be performed to verify compliance with the cybersecurity and security controls requirements of 3.11 to include all subparagraphs. Contractor and/or manufacturer certification may be used in lieu of or to augment the analysis, where applicable, to verify compliance.

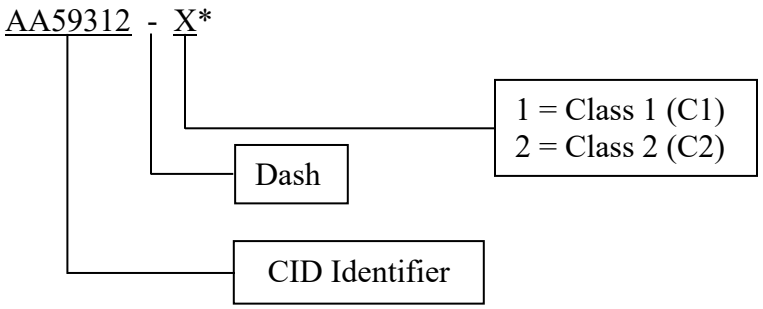
6. **PACKAGING**.

6.1 Preservation and packaging. Preservation, packing, and marking shall be as specified in the contract or order.

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7. **NOTES**

7.1 Part or Identifying Number (PIN). The PIN is constructed as follows.



*This example (AA59312-C1) describes a PIN for CID A-A-59312.

7.2 Source of documents.

7.2.1 Air Force Instructions (AFI). Copies of AFI's are available online at https://www.e-publishing.af.mil/.

7.2.2 Department of Defense and Federal documents. Department of Defense and Federal documents are available online at https://quicksearch.dla.mil/.

7.2.3 Department of Defense Instructions (DoDI). Copies of DoDI's are available online at https://www.esd.whs.mil/DD/.

7.2.4 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

7.2.5 ASTM documents. Copies of ASTM documents may be obtained from the ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, or online at https://www.astm.org/.

7.2.6 American Welding Society (AWS) documents. Copies of AWS documents may be obtained from the American Welding Society, 8669 Northwest 36 Street, Suite 130, Miami, FL 33166-6672, or online at https://www.aws.org/w/a/.

7.2.7 Cortec documents. Copies of Cortec documents may be obtained from Cortec Corporation, 4119 White Bear Parkway, St. Paul, MN 55110, or online at https://www.cortecvci.com.

7.2.8 Code of Federal Regulations (CFR). Copies of CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at https://www.ecfr.gov/cgi-bin/ECFR?page=browse.

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7.2.9 NAS standards. Copies of SAE standards may be obtained from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington VA 22209, or online at https://www.aia-aerospace.org/.

7.2.10 National Electrical Manufactures Association (NEMA) documents. Copies of NEMA documents may be obtained from the National Electrical Manufactures Association, 1300 North 17th Street, Suite 900, Arlington, VA 22209, or online at https://www.nema.org.

7.2.11 Society of Automotive Engineers (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at https://www.sae.org/.

7.2.12 Tire and Rim Association (TRA) standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321, or online at https://www.us-tra.org/publications.html.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID

b. Packaging requirements (see 6.1)

c. Finish color required (gloss white) (see 3.2.1.2)

d. Finish color required (Coast Guard Blue) (see 3.2.1.2)

e. Temporary CPC (see 3.2.1.4)

f. Winterization A (see 3.3.1, 3.3.2, and 3.7.6

g. Class 2 four-wheel drive FLTT (see 3.6.4)

7.4 National Stock Number (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN		Class	PIN
1740-01-068-8945		1	AA59312-C1
1740-01-452-4117		2	AA59312-C2

7.5 Definitions.

7.5.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open-end, boxed-end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex-key wrenches, in SAE sizes up to and including 1.00 inch and metric sizes up to and including 25 mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

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7.5.2 Special tool. A tool that is not commercially and readily available from a source other than the FLTT contractor.

7.6 Additional information.

7.6.1 Engine waiver. Tier II "on-highway" represents the highest level of exhaust and evaporative emissions standards that will permit the use of high sulfur fuels. The Air Force possesses an Environmental Protection Agency (EPA) waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a tag or decal for the engine stating "[insert engine description here] – This engine has an exemption for national security under 40 CFR 89.908 and/or 40 CFR 1068.225".

7.7 Key words.

Aircraft parking
FLTT
Tow tractor
Tug

7.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due the extent of the changes.

Concluding Material:

Custodian:
Air Force - 184
Army - AV

Preparing activity:
Air Force - 184

Agent:
Air Force - 110
(Project 1740-2020-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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